

AI in Fluids: Week 16

Supersonic Shape Optimization and Sonic-Boom Physics

From compressible-flow theory to verified CFD, POD surrogates, and design recomputation

The learning question. How can we reshape a supersonic body so that its off-body pressure signature is reduced, use machine learning to search the design space efficiently, and still make only claims that are supported by verified CFD?

After Week 16, you should be able to

1. Explain the speed of sound, Mach number, Mach angle, compression waves, expansion fans, shocks, and the origin of a sonic-boom pressure signature.
2. Distinguish near-field pressure shaping from atmospheric propagation and perceived ground loudness.
3. Derive the fixed-volume two-parameter geometry used in the teaching problem.
4. Write the axisymmetric Euler equations used by the CFD solver and explain the pressure coefficient and pressure-drag objective.
5. Distinguish iterative convergence, mesh sensitivity, analytical verification, physical checks, experimental validation, and ML validation.
6. Explain POD compression, training/validation/test separation, and why extrapolation is harder than interpolation.
7. Formulate the constrained design problem, interpret the surrogate proposal, and accept a design only after fresh CFD recomputation.
8. Reproduce the principal numerical evidence and state precisely what Week 16 does and does not demonstrate.

The lecture follows the same teaching philosophy as Week 1: begin with the governing physics, derive the equations, connect each equation to the numerical implementation, inspect the actual computational evidence, and end with the limits of the conclusion.

Companion material. The executable notebook, CFD reproduction guide, and retained numerical evidence are all distributed in the Week 16 folders of the FlowMLLab repository. The exact repository paths are listed in the References section.

Sections	Learning route
1–3	Supersonic information propagation, shocks/expansions, and sonic-boom signatures
4–5	Near-field pressure, ground propagation, and loudness
6–8	Geometry, axisymmetric Euler CFD, numerical acceptance, and verification
9–10	Dataset generation, POD, neural surrogates, and error metrics
11–12	Constrained optimization, fresh CFD verification, and off-design checks
13–14	NASA reference extension, failed reproduction, and validation hierarchy
15	Reproducible campaign record, exercises, and take-away

Claim boundary. The accepted Week 16 result is an axisymmetric, zero-incidence, inviscid teaching problem at $M_\infty = 1.8$. The primary quantity is the near-field pressure signature at $r/L = 0.5$. Atmospheric propagation and PLdB are not computed. A smaller near-field peak is therefore not, by itself, proof of a quieter ground sonic boom.

1 Why a supersonic body creates a pressure signature

1.1 Speed of sound and Mach number

For a calorically perfect gas, small pressure disturbances travel at the local speed of sound

$$a = \sqrt{\gamma RT}, \quad (1)$$

where γ is the specific-heat ratio, R is the gas constant, and T is absolute temperature. The Mach number is

$$M = \frac{U}{a}. \quad (2)$$

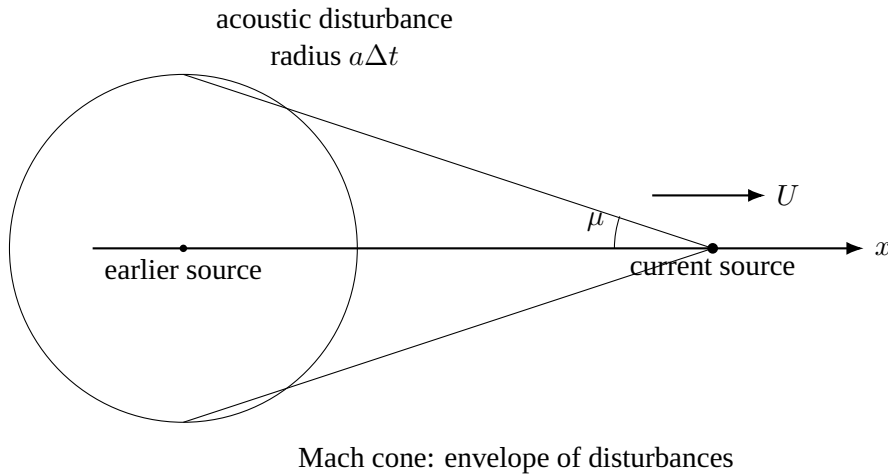
For $M < 1$, acoustic information can propagate upstream. For $M > 1$, the body moves faster than small disturbances can travel upstream, so disturbances are confined to a downstream cone.

1.2 The Mach angle

Consider a disturbance emitted at one instant. After a time Δt , the disturbance has expanded a distance $a\Delta t$, while the source has moved $U\Delta t$. The envelope of all such disturbance spheres defines the Mach cone. Geometry gives

$$\sin \mu = \frac{a}{U} = \frac{1}{M}, \quad \mu = \sin^{-1}\left(\frac{1}{M}\right). \quad (3)$$

At $M = 1.8$, $\mu \approx 33.75^\circ$.



Physical meaning. Increasing M narrows the cone. This changes where pressure disturbances intersect an off-body observation line and how signals from different parts of the vehicle interact.

From theory to our CFD. The clean Week 16 campaign fixes $M_\infty = 1.8$. Later, the retained baseline and optimized geometries are recomputed at $M = 1.7$ and 1.9 to test whether the design improvement survives a modest change in operating condition.

Check your understanding. Compute μ at $M = 1.2$ and $M = 3$. Which case has the wider region of upstream influence?

2 Compression waves, shocks, and expansion fans

A supersonic body turns the flow. Turning the flow into itself produces compression; turning it away produces expansion. The local slope distribution of the body therefore generates a train of compression and expansion disturbances.

2.1 Oblique compression and shock formation

For a two-dimensional wedge or a locally inclined slender surface, the oblique-shock relation connects the upstream Mach number M_1 , shock angle β , and flow deflection angle θ :

$$\tan \theta = 2 \cot \beta \frac{M_1^2 \sin^2 \beta - 1}{M_1^2 (\gamma + \cos 2\beta) + 2}. \quad (4)$$

The normal component is $M_{n1} = M_1 \sin \beta$. Across the shock,

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_{n1}^2 - 1). \quad (5)$$

A weak compression wave travels slightly faster in its higher-pressure region. A sequence of compressions can therefore steepen and coalesce into a shock.

2.2 Prandtl–Meyer expansion

A smooth convex turn produces a centered expansion fan. The Prandtl–Meyer function is

$$\nu(M) = \sqrt{\frac{\gamma + 1}{\gamma - 1}} \tan^{-1} \sqrt{\frac{\gamma - 1}{\gamma + 1} (M^2 - 1)} \quad (6)$$

$$- \tan^{-1} \sqrt{M^2 - 1}. \quad (7)$$

For an isentropic turn through angle θ ,

$$\theta = \nu(M_2) - \nu(M_1). \quad (8)$$

Pressure falls through the fan while Mach number rises.

2.3 Why body shape matters

A slender body's axial radius or area distribution controls where the local surface turns compress or expand the flow. Changing the distribution can split a strong compression into weaker events or shift wave interactions. Low-boom design is therefore not merely “make the nose sharper”: the full longitudinal volume and lift distribution matters.

Check your understanding. Why can two bodies with the same total volume generate different off-body pressure signatures?

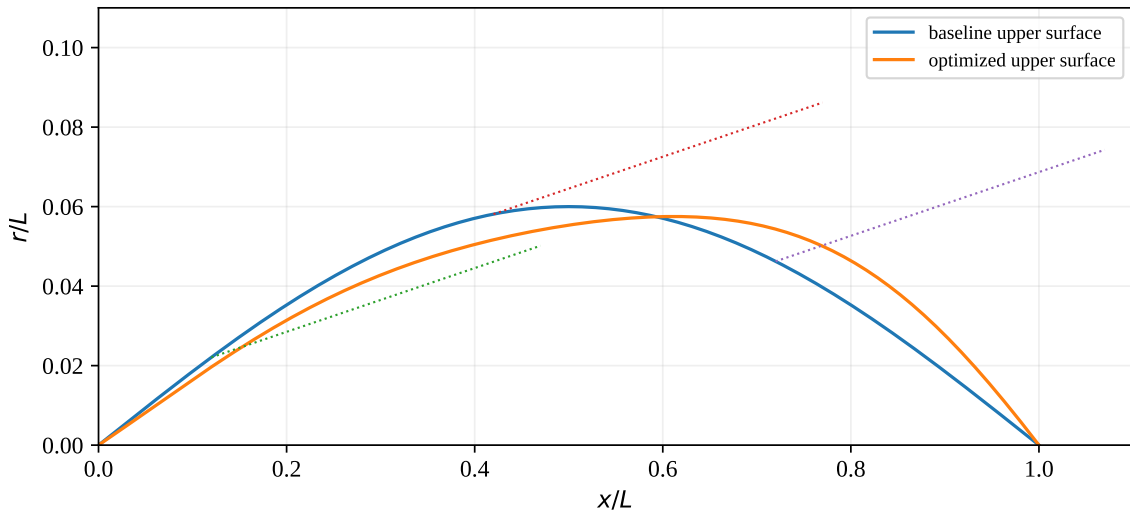
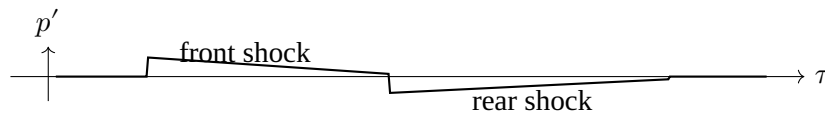


Figure 1: Teaching-body profiles with conceptual supersonic wave directions. The dotted rays are pedagogical; the actual CFD solves the nonlinear Euler equations.

3 From many waves to a sonic-boom signature

3.1 Near-field and far-field waveforms

Immediately around a vehicle, the pressure field contains multiple shocks and expansions associated with nose, volume changes, lift-producing surfaces, propulsion integration, and the tail. During propagation, nonlinear steepening, geometric spreading, and atmospheric absorption alter the signal. For a conventional configuration, multiple features may merge into an approximately N-shaped waveform with a rapid positive jump, gradual decay, a negative excursion, and a final recovery.



3.2 Equivalent area and the Whitham idea

In classical slender-body sonic-boom theory, the aircraft is represented through an equivalent area distribution $A_e(x)$ that combines volume and, for lifting configurations, a lift-equivalent contribution. A commonly used Whitham-type transform has the form

$$F(y) \propto \int_0^y \frac{A_e''(x)}{\sqrt{y-x}} dx, \quad (9)$$

with normalization depending on the chosen convention. The key lesson is that the pressure signature depends strongly on the *second axial variation* of equivalent area, not merely on total volume. Smoothing or redistributing area changes where compression and expansion features appear.

Why we do not use the F -function as the numerical solver here. The Week 16 teaching case computes the nonlinear axisymmetric Euler field directly with SU2. The equivalent-area viewpoint is used to explain why shape redistribution can alter the pressure signature and why fixed volume is a meaningful design constraint.

Check your understanding. If two bodies have equal volume but different $A_e''(x)$, should they have identical boom signatures? Explain.

4 What pressure quantity are we optimizing?

CFD commonly reports the pressure coefficient

$$C_p = \frac{p - p_\infty}{q_\infty}, \quad q_\infty = \frac{1}{2} \rho_\infty U_\infty^2. \quad (10)$$

For a calorically perfect gas,

$$\rho_\infty U_\infty^2 = \gamma p_\infty M_\infty^2, \quad (11)$$

so pressure excess normalized by freestream static pressure is

$$\frac{\Delta p}{p_\infty} = \frac{\gamma M_\infty^2}{2} C_p. \quad (12)$$

At $M_\infty = 1.8$ and $\gamma = 1.4$, the conversion factor is 2.268. At $M = 1.6$, it is 1.792.

This conversion is essential when comparing sources: NASA sonic-boom records may use $\Delta p/p_\infty$, whereas the teaching-design plots use C_p .

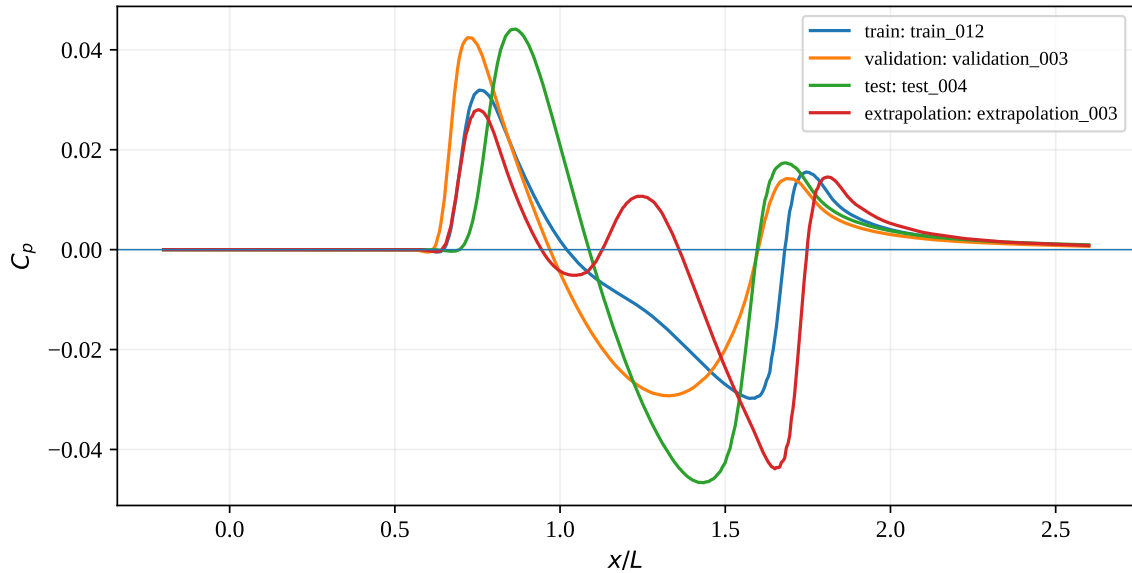


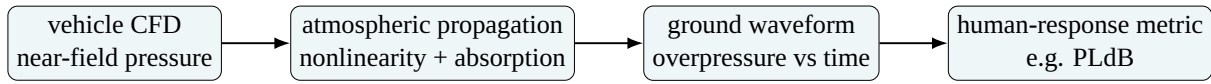
Figure 2: Representative accepted SU2 8.0.1 near-field pressure signatures from each data split. The waveform structure, not only the scalar peak, is important.

From theory to our CFD. The primary design objective is the maximum positive C_p along the extraction line $r/L = 0.5$. Pressure signatures are also extracted at $r/L = 0.25$ and 0.75 for diagnostic comparisons.

Check your understanding. Convert $C_p = 0.03$ to $\Delta p/p_\infty$ at $M = 1.8$.

5 Near-field shaping is not the same as ground loudness

A complete sonic-boom prediction has at least three stages:



5.1 Generalized Burgers-type propagation

A schematic weak-shock propagation equation can be written as

$$\frac{\partial p'}{\partial s} + \underbrace{\beta_n p' \frac{\partial p'}{\partial \tau}}_{\text{nonlinear steepening}} = \underbrace{\delta \frac{\partial^2 p'}{\partial \tau^2}}_{\text{thermoviscous absorption}} + \underbrace{\mathcal{G}[p']}_{\text{geometric spreading}} + \underbrace{\mathcal{A}[p']}_{\text{atmospheric effects}}, \quad (13)$$

where s is propagation distance and τ is retarded time. Practical augmented Burgers solvers can also include molecular relaxation, humidity, temperature stratification, wind, and ray-tube geometry.

5.2 Why PLdB requires more than a peak

Perceived loudness depends on the full time history and its frequency content. Two waveforms with the same maximum overpressure can have different rise times, durations, spectral distributions, and therefore different loudness. A smaller near-field C_p peak is a useful design indicator, but it is not a PLdB calculation.

Claim boundary. Week 16 stops before atmospheric propagation. The correct conclusion is “verified reduction of a near-field pressure peak under the stated CFD model,” not “verified quieter sonic boom on the ground.”

6 The teaching geometry: changing shape without shrinking the body

Let $s = x/L$ and define an unnormalized radius function

$$f(s; a, b) = \sin(\pi s) \exp[a(2s - 1) + b \cos(2\pi s)]. \quad (14)$$

The radius is

$$r(s) = c f(s; a, b). \quad (15)$$

For a body of revolution,

$$V = \pi L \int_0^1 r(s)^2 ds = \pi L c^2 \int_0^1 f(s)^2 ds. \quad (16)$$

Solving for c gives

$$c = \sqrt{\frac{V}{\pi L \int_0^1 f(s)^2 ds}}. \quad (17)$$

Thus every (a, b) pair has the same target volume. The teaching case uses $L = 1$ m and

$$V = \frac{\pi(0.06 \text{ m})^2 L}{2}. \quad (18)$$

Parameter a shifts volume fore or aft; parameter b changes the balance between central and end regions. Fixed volume prevents a trivial optimizer from reducing pressure simply by making the body smaller.

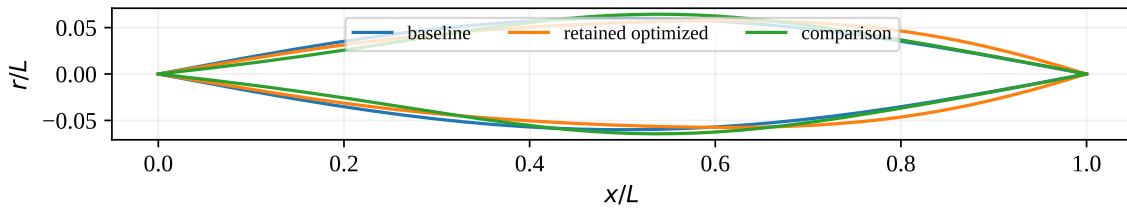


Figure 3: Three fixed-volume bodies. The retained optimized design is $a = 0.32495$, $b = 0.12238$.

What this geometry is not. It is not the Beihang TMS-10 aircraft, not a wing-body configuration, and not a complete transport vehicle. Cabin volume, structural thickness, lift, trim, propulsion, and viscous drag are outside this teaching model.

Check your understanding. Why is fixed volume necessary for a meaningful comparison? What additional constraints would a transport-aircraft design require?

7 Governing equations used in the CFD

7.1 Axisymmetric compressible Euler equations

The computational mesh is two-dimensional in the meridional (x, r) plane, but the physical body is three-dimensional and axisymmetric. A convenient weighted conservative form is

$$\frac{\partial(rU)}{\partial t} + \frac{\partial(rF)}{\partial x} + \frac{\partial(rG)}{\partial r} = S, \quad (19)$$

with

$$U = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \quad F = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (\rho E + p)u \end{bmatrix}, \quad G = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (\rho E + p)v \end{bmatrix}, \quad (20)$$

and the axisymmetric geometric source

$$S = \begin{bmatrix} 0 \\ 0 \\ p \\ 0 \end{bmatrix}. \quad (21)$$

The pressure is recovered from total energy using the perfect-gas relation

$$p = (\gamma - 1) \left[\rho E - \frac{\rho(u^2 + v^2)}{2} \right]. \quad (22)$$

Why the radial source matters. A planar two-dimensional Euler calculation omits the geometric divergence of an axisymmetric flow. Treating the meridional outline as an ordinary 2-D airfoil would therefore solve a different physical problem.

7.2 Pressure drag

For an axisymmetric body with radius $r(x)$, the axial pressure force can be written

$$D_p = 2\pi \int_0^L (p - p_\infty) r \frac{dr}{dx} dx, \quad (23)$$

with sign determined by the surface orientation convention. The nondimensional teaching metric is

$$C_{D,p} = \frac{D_p}{q_\infty L^2}. \quad (24)$$

This is pressure drag only. Skin friction is absent because the Euler equations are inviscid.

From theory to our CFD. The design objective minimizes the maximum C_p at $r/L = 0.5$ subject to $C_{D,p} \leq 1.02C_{D,p,0}$ and the exact fixed-volume parameterization.

8 What we actually ran: CFD setup and acceptance gates

8.1 Physical and numerical setup

The accepted clean campaign uses Gmsh meshes and the checksum-pinned official SU2 8.0.1 solver. The freestream and geometry are deliberately simple so that every design can be recomputed and audited.

Quantity	Teaching value
Geometry	axisymmetric body of revolution, zero incidence
Length	$L = 1$ m
Freestream Mach	$M_\infty = 1.8$
Freestream pressure	$p_\infty = 101325$ Pa
Freestream temperature	$T_\infty = 288.15$ K
Governing model	compressible Euler, ideal gas
Primary extraction line	$r/L = 0.5$
Other extraction lines	$r/L = 0.25$ and 0.75
Reference area for $C_{D,p}$	L^2
Training mesh level	1.5
Typical design cells	45,408 (level 1.5), 80,640 (level 2)

8.2 Why a residual is not enough

Every accepted label must pass numerical and full-field physical rejection gates. The clean campaign requires finite positive pressure and density, final $\log_{10}$ density residual no larger than -9 , at least five orders of residual reduction, pressure-drag variation below 10^{-4} over the final 100 iterations, maximum total-enthalpy deviation no larger than 10%, and density below 110% of the isentropic stagnation bound. These thresholds are rejection criteria, not estimates of physical uncertainty.

8.3 Accepted CFD field evidence

Check your understanding. A run reaches a density residual of 10^{-11} but has a large local total-enthalpy defect near the tip. Should it be accepted? Why not?

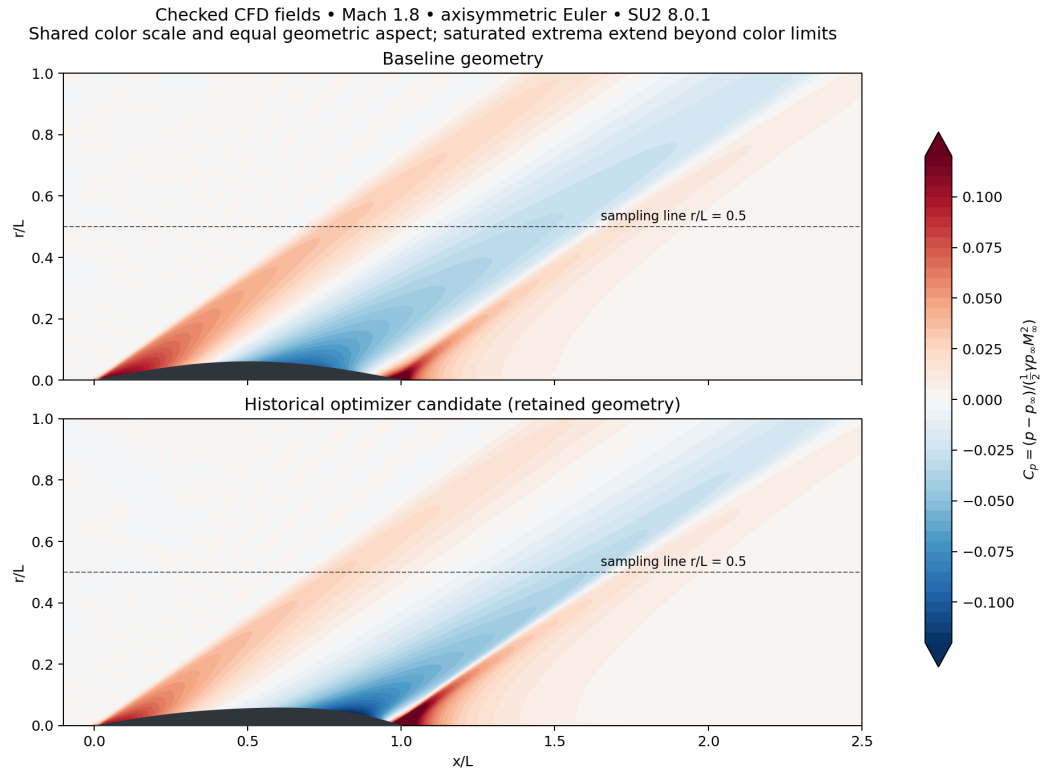


Figure 4: Accepted SU2 8.0.1 fields for the teaching body. The field plots are part of the retained physical/numerical evidence, not decorative illustrations.

9 Verification before machine learning

Verification asks: are we solving the chosen equations correctly? It is different from validation, which asks whether the model agrees with reality for the intended quantity.

9.1 Taylor–Maccoll cone benchmark

Supersonic flow over a cone admits an axisymmetric similarity solution. The Taylor–Maccoll ODE system is integrated from an attached conical shock to the cone surface. The shock angle is adjusted until the polar velocity satisfies the wall-tangency condition. This provides an independent pressure reference that does not use SU2’s finite-volume discretization.

For a 7-degree cone at $M = 1.8$, the retained mesh study uses 24,000, 96,000, and 216,000 fluid cells. The wall- C_p errors relative to the independent Taylor–Maccoll solution are 4.563%, 2.156%, and 1.162%. The last-two mean- C_p change is 1.006%.

Fluid cells	CFD mean C_p	error to Taylor–Maccoll	numerical/physical gates
24,000	0.059175	4.563%	pass
96,000	0.060667	2.156%	pass
216,000	0.061284	1.162%	pass

Interpretation. The cone verifies a specific component of the solver/model chain: axisymmetric inviscid supersonic wall pressure. It does not experimentally validate the boom waveform, the teaching geometry, or the neural network.

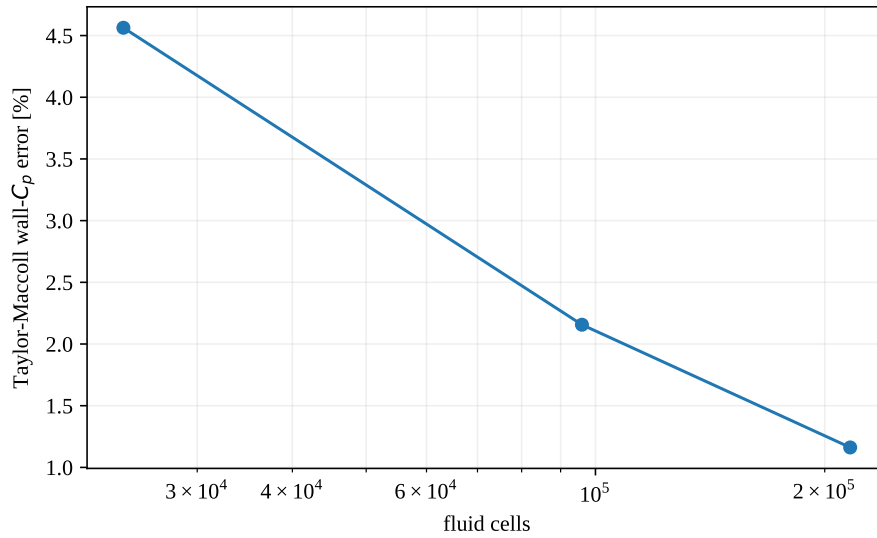


Figure 5: Taylor–Maccoll wall-pressure error decreases with refinement. The coarse pressure criterion fails, while the declared refined-family verification criteria pass.

10 The 44-case dataset and the reduced-order learning problem

10.1 Frozen data split

The clean SU2 8.0.1 campaign contains 44 geometries: 24 training, 6 validation, 8 test, and 6 extrapolation cases. All belong to the same two-parameter family. The extrapolation cases extend one shape parameter beyond the training region; they do not introduce a new topology.

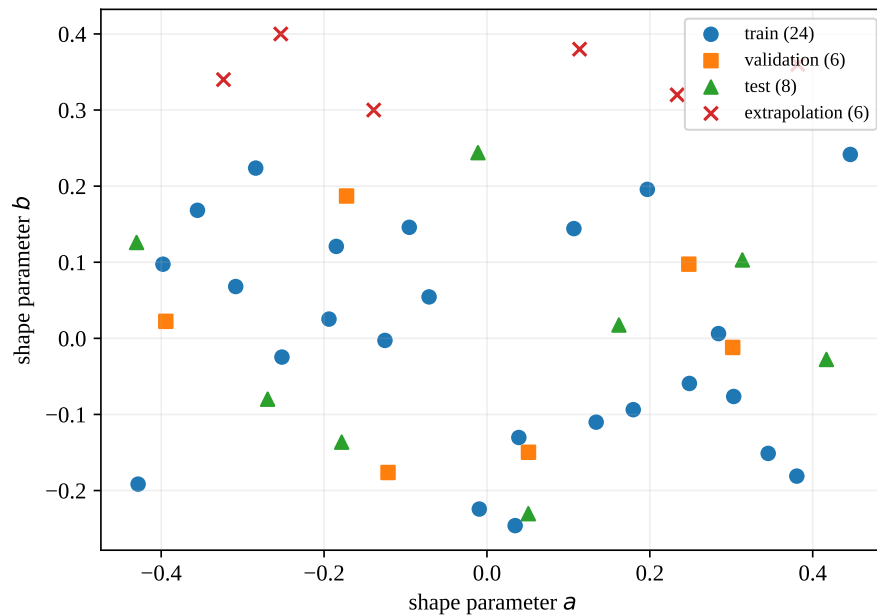


Figure 6: Frozen geometry split. Training statistics, scalars, and POD modes must be fitted only on the training subset.

10.2 Proper Orthogonal Decomposition

Let each CFD signature be a row vector $y_i = [C_p(x_1), \dots, C_p(x_m)]$. Form the training matrix Y and subtract the training mean:

$$Y' = Y - \bar{Y}. \quad (25)$$

The singular value decomposition is

$$Y' = U \Sigma V^T. \quad (26)$$

The first K rows of V^T are POD modes $\phi_k(x)$. A waveform is reconstructed as

$$C_p(x; \theta) \approx \bar{C}_p(x) + \sum_{k=1}^K z_k(\theta) \phi_k(x). \quad (27)$$

The retained model uses $K = 12$ full-SVD modes. The retained energy is

$$E_K = \frac{\sum_{k=1}^K \sigma_k^2}{\sum_j \sigma_j^2}. \quad (28)$$

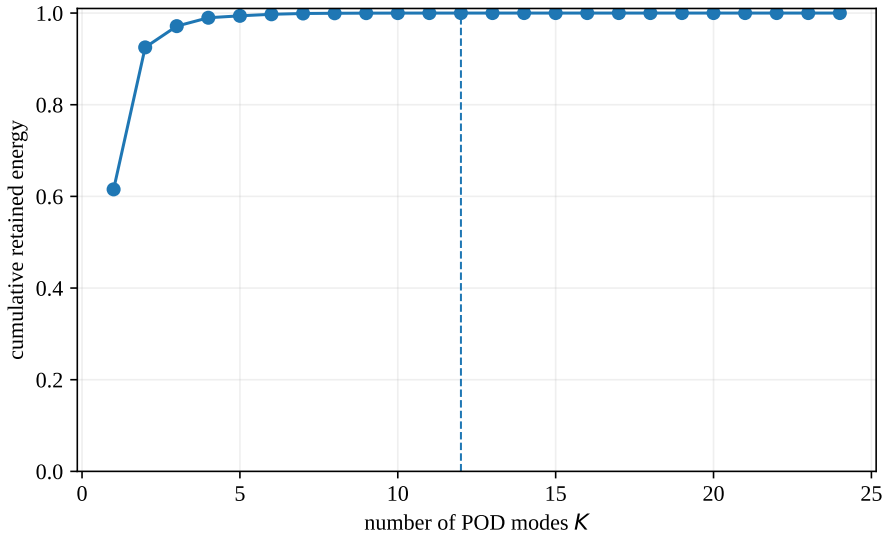


Figure 7: Cumulative POD energy of the 24 training waveforms. Compression reduces a 561-point CFD signature to a small set of modal coefficients.

11 Neural surrogate: what is learned, and how do we judge it?

The retained clean model maps two geometry parameters to 12 POD coefficients and log pressure drag:

$$(a, b) \longrightarrow [z_1, \dots, z_{12}, \log C_{D,p}]. \quad (29)$$

Its architecture is a 32–32 tanh MLP trained with L-BFGS, $\alpha = 0.01$, random seed 16, and training-only scaling. The full-SVD POD basis is also fitted only on the 24 training cases.

11.1 Three error measures

Whole-waveform relative error:

$$\epsilon_{\text{wave}} = \frac{\|\hat{y} - y\|_2}{\|y\|_2}. \quad (30)$$

Mean positive-peak relative error:

$$\epsilon_{\text{peak}} = \left| \frac{\max \hat{C}_p - \max C_p}{\max C_p} \right|. \quad (31)$$

Pressure-drag relative error:

$$\epsilon_D = \left| \frac{\hat{C}_{D,p} - C_{D,p}}{C_{D,p}} \right|. \quad (32)$$

A shock can be shifted in x while retaining a similar peak amplitude; therefore peak error alone can hide a poor waveform.

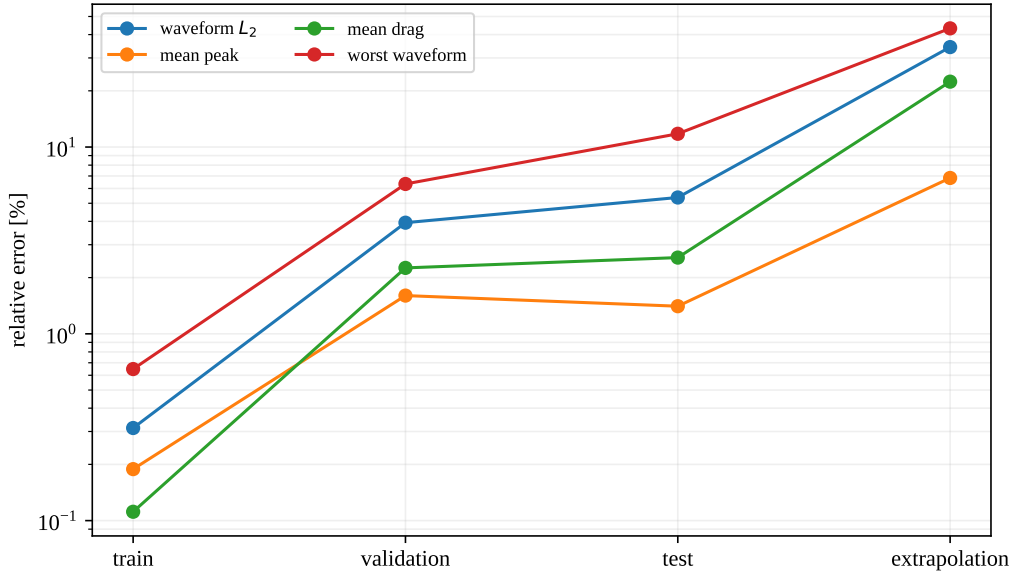


Figure 8: Clean-model errors across the frozen splits. Extrapolation is substantially harder than interpolation. A log scale is used so that training and extrapolation errors remain visible together.

Split	wave L_2	mean peak	mean drag	worst waveform
Train (24)	0.313%	0.189%	0.112%	0.648%
Validation (6)	3.935%	1.602%	2.254%	6.343%
Test (8)	5.369%	1.406%	2.559%	11.775%
Extrapolation (6)	34.286%	6.831%	22.389%	43.190%
Finer CFD test (8)	7.204%	2.237%	2.274%	15.372%

Lesson. Aggregate criteria can pass while one geometry has a much larger error. Report the worst individual case, and do not claim reliable extrapolation from good interpolation results.

12 Optimization: let the surrogate propose, let CFD decide

The constrained design problem is

$$\min_{a,b} \max_x C_p(x, r/L = 0.5; a, b) \quad (33)$$

subject to

$$C_{D,p}(a, b) \leq 1.02C_{D,p,0}, \quad (34)$$

and the exact fixed-volume parameterization of Section 6.

A learned surrogate makes a large number of candidate evaluations cheap, but this creates a new danger: an optimizer actively searches for regions where surrogate error can be exploited. Ordinary test-set accuracy is therefore not enough. The proposed optimum must be meshed again and solved with CFD.

12.1 Retained design

The retained candidate has

$$a = 0.3249533, \quad b = 0.1223779. \quad (35)$$

At mesh level 1.5, the baseline and optimized bodies use 45,408 fluid cells. At level 2 they use 80,640 cells.

Case	peak C_p	$C_{D,p}$	peak change	drag change
Baseline, level 1.5	0.0333075	0.00135423	—	—
Optimized, level 1.5	0.0262504	0.00130281	−21.188%	−3.797%
Baseline, level 2	0.0341276	0.00136763	—	—
Optimized, level 2	0.0270753	0.00131947	−20.665%	−3.522%

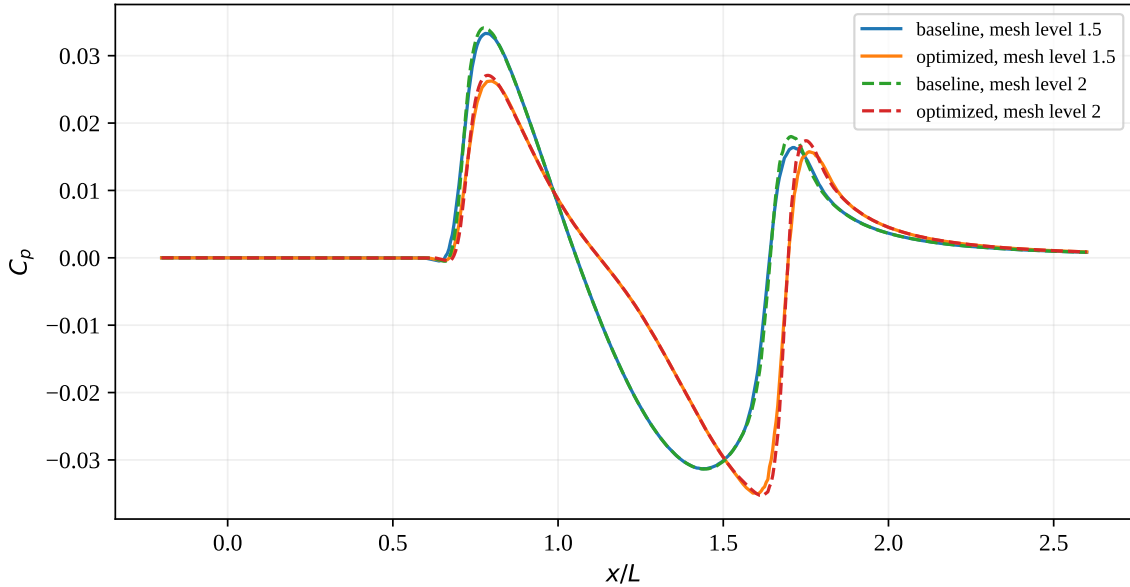


Figure 9: Fresh SU2 8.0.1 recomputation of baseline and retained optimized signatures on two mesh levels. The improvement is measured from CFD, not accepted from the neural prediction.

12.2 Mesh sensitivity

For the baseline, the level-1.5 to level-2 changes are 2.403% in peak, 0.980% in pressure drag, and 3.586% in the waveform L_2 measure. For the optimized body they are 3.047%, 1.262%, and 5.395%. The declared peak and drag gates pass. The waveform difference is reported separately rather than hidden behind scalar agreement.

Check your understanding. Why is a fresh CFD solve of the optimized geometry more important than a very small training error?

13 Off-design behavior and robustness

The surrogate was trained at $M = 1.8$. To test the retained geometry rather than the learned model, the baseline and optimized bodies were recomputed directly with SU2 at $M = 1.7$ and 1.9 .

Mach	peak-pressure reduction	pressure-drag reduction
1.7	21.010%	3.231%
1.8	21.188%	3.797%
1.9	21.166%	4.311%

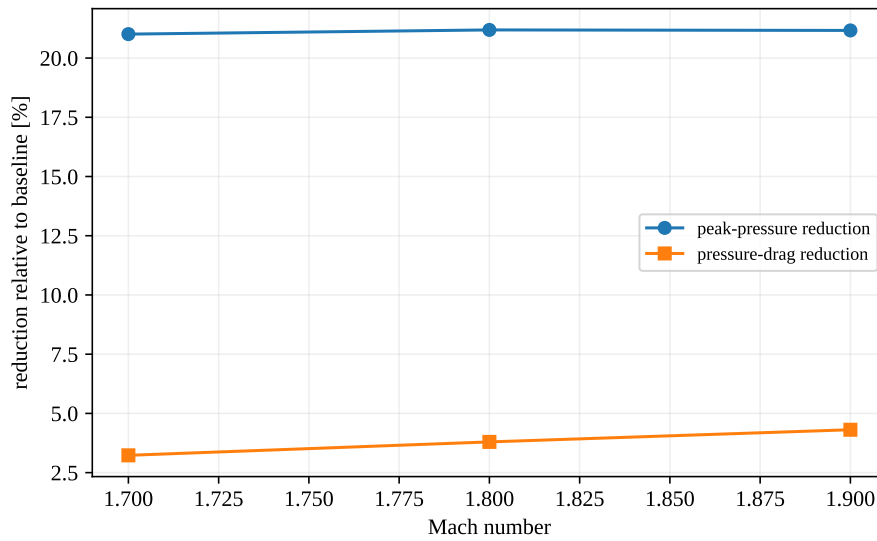


Figure 10: Direct CFD off-design evidence. These points are not neural-network predictions.

What this establishes. The local design benefit survives two nearby Mach-number perturbations for the teaching geometry and the chosen metrics. It does not establish a complete flight envelope, uncertainty quantification, or robustness to altitude, angle of attack, Reynolds number, atmospheric state, or manufacturing variation.

14 NASA SEEB-ALR: a useful failed research extension

The NASA SEEB-ALR benchmark is separate from the two-parameter teaching body. It provides an as-built geometry and off-body wind-tunnel records at $M = 1.6$, making it attractive for experimental validation of a CFD chain.

14.1 Pressure convention and geometry alignment

The NASA pressure ordinate is $\Delta p/p_\infty$, not C_p . At $M = 1.6$ and $\gamma = 1.4$, use

$$\frac{\Delta p}{p_\infty} = 1.792 C_p. \quad (36)$$

The retained source macros prescribe longitudinal shifts; no fitted shock-position shift should be introduced merely to reduce error. The original finite nose and downstream sting must be retained.

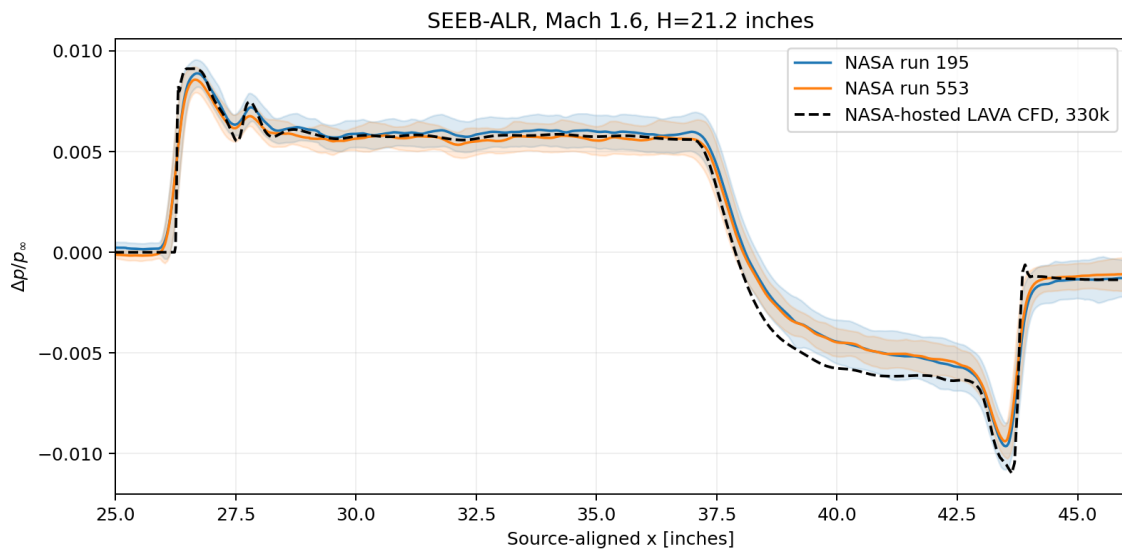


Figure 11: Retained NASA-hosted LAVA CFD compared with two NASA wind-tunnel records. This is a reference comparison, not a successful SU2 validation by this project.

For the fixed 25–46 inch comparison window, the retained LAVA-to-experiment waveform errors are 14.75% and 15.02%, with peak errors 2.64% and 6.41% for NASA runs 195 and 553, respectively.

14.2 Why our attempted SU2 reproduction was rejected

The NASA recovery runs did not establish an accepted physically checked, mesh-refined family. Earlier nose treatment produced unphysical stagnation behavior. Later replays still failed convergence/physical gates. The correct scientific action is to preserve the failure, document it, and exclude it from the accepted release claims.

Claim boundary. A solver exit code, a visually plausible pressure curve, or a small residual cannot turn an unaccepted NASA run into validation evidence. The Week 16 release therefore reports NASA as a deferred failed research extension.

15 Verification, validation, and ML validation are different

Question	Typical evidence	Week 16 example
Numerical verification	residuals, mesh refinement, analytical benchmark, conservation/physical checks	Taylor–Maccoll cone; design mesh sensitivity; full-field gates
Physical validation	comparison with suitable experiment and uncertainty	NASA SEEB-ALR is a target, but the SU2 reproduction is not accepted
ML validation	frozen data splits, train-only pre-processing, held-out CFD, worst-case errors	clean 24/6/8/6 split; finer-CFD comparison; extrapolation stress test
Design verification	fresh high-fidelity recomputation of the proposed optimum	baseline/optimized SU2 8.0.1 recomputation on two meshes

The central logical chain is

physics → verified CFD → reduced model → surrogate proposal → fresh CFD check.

The neural model accelerates the search; it does not replace verification.

15.1 What we actually completed

Computational campaign	Outcome	Retained evidence
44-case clean-label SU2 8.0.1 campaign	PASS	24 train / 6 validation / 8 test / 6 extrapolation; all declared full-field gates pass
Fixed clean-label MLP audit	PASS	same-mesh and finer-mesh metrics retained; extrapolation failure reported
Taylor–Maccoll three-mesh refinement	PASS	finest wall- C_p error 1.162%; last-two change 1.006%
Retained optimized-design recomputation	PASS	peak reduction 21.188% and 20.665% on two meshes; drag constraint passes
Off-design $M = 1.7, 1.9$ recomputation	PASS	near-field peak reduction remains about 21%
NASA SEEB-ALR SU2 reproduction	FAIL / deferred	failure retained; excluded from accepted validation claims

GitHub Actions provenance for the accepted scientific runs. Clean campaign run 35944299803; clean model run 35944888983; retained-design verification run 35943668623; Taylor–Maccoll refinement run 35943668572; verified publication candidate run 36002078532. NASA recovery/replay runs 35944409924 and 35946027208 failed and remain deferred.

16 Student interpretation and exercises

Exercise 1: Mach geometry. Compute μ for $M = 1.4, 1.6, 2.0, 2.5$. Explain why the cone narrows as Mach increases.

Exercise 2: pressure normalization. At $M = 1.8$, convert $C_p = 0.03$ to $\Delta p/p_\infty$. Repeat at $M = 1.6$.

Exercise 3: fixed volume. Choose two new (a, b) pairs, evaluate $r(s)$ numerically, and verify the target volume to better than 10^{-5} relative error.

Exercise 4: POD truncation. Repeat the learning problem with $K = 4, 8, 12$. Compare retained energy and test waveform error.

Exercise 5: data leakage. Intentionally fit the POD basis on all 44 cases. Why might the apparent test error improve, and why is that result scientifically invalid?

Exercise 6: shock location versus peak. Find the test case with the largest waveform L_2 error. Determine whether the dominant error is amplitude, shock position, or both.

Exercise 7: tighter constraint. Replace the drag limit $1.02C_{D,p,0}$ with $1.00C_{D,p,0}$. How does the feasible design region change?

Exercise 8: claim discipline. Write one sentence that is justified by the Week 16 evidence and one sentence that would be an overclaim. Your justified sentence must distinguish near-field pressure from ground loudness.

Check your understanding. If the optimized near-field peak falls 21% but the atmospheric propagation model is never run, what is the strongest defensible statement?

17 Take-away

The most important result of Week 16 is not a particular pair (a, b) . It is a defensible computational workflow:

1. start from compressible-flow physics and meaningful constraints;
2. solve a clearly stated CFD model;
3. verify the solver/model chain with independent numerical evidence;
4. generate a frozen training/validation/test dataset;
5. compress the waveform with training-only POD;
6. use a surrogate to explore designs cheaply;
7. recompute the proposed design with CFD;
8. report mesh sensitivity, worst-case ML errors, off-design checks, and failed extensions;
9. stop the scientific claim at the boundary of the evidence.

For the retained teaching body, the optimized geometry reduces the verified near-field peak by about 21% while also reducing pressure drag on two meshes. The clean neural surrogate is accurate for interpolation within the two-parameter family but degrades strongly under extrapolation. NASA experimental CFD validation remains unresolved. Ground sonic-boom loudness remains outside the computed scope.

References

Zheng, Q., Liang, Y., Yang, Y. and Pan, C. (2026). “Research on low-drag low-boom supersonic transport configuration using an MDO framework and deep learning methods.” *Aerospace Science and Technology*, 178, 113218. DOI: 10.1016/j.ast.2026.113218.

Taylor, G. I. and Maccoll, J. W. (1933). “The air pressure on a cone moving at high speeds.” *Proceedings of the Royal Society A*, 139, 278–297.

SU2 project documentation and governing-equation references: <https://su2code.github.io/>. The accepted Week 16 clean campaign uses the checksum-pinned official SU2 8.0.1 executable.

Gmsh reference manual: <https://gmsh.info/doc/texinfo/gmsh.html>.

AIAA/NASA Sonic Boom Prediction Workshop material: <https://lbpw.larc.nasa.gov/>. NASA SEEB-ALR source data and original coordinate macros are retained in the Week 16 reference directory with provenance records.

FlowMLLab Week 16 evidence is retained under the repository paths `results/week16_lowboom/`, `notebooks/week16/`, and `cases/week16_lowboom/`. The model and NASA validation guides are in the Week 16 notebook directory.